

Closing the gaps: a complete proof that $\mathcal{P}^{[\deg \mathfrak{n}]}$ is unramified at $\eta_{\mathfrak{n}}$ for $G = \mathbb{Z}/p^r\mathbb{Z}$

Working note

Abstract

The note *Completing the induction* proves the theorem “ $\mathcal{P}^{[\deg \mathfrak{n}]}$ is unramified at $\eta_{\mathfrak{n}}$ for $G = \mathbb{Z}/p^r\mathbb{Z}$ ” *conditionally* on two unproven steps (Gaps 1 and 2 of the review): (1) that the Witt datum of the torsor at p_{∞} may be taken to be a *reduced vector of polynomials in x^{-1}* satisfying the pole bound $p^{r-1-j}m_j < d$, and (2) that the symmetric-power cover at η_C is the Artin–Schreier–Witt cover of the Witt sum $\bigoplus_{i=1}^d \tilde{f}(x_i)$. This note supplies both missing results. Gap 1 is closed by a W_r -analogue of the thesis’s realization lemma over $\mathbb{G}_m$: every class in $W_r(k((x)))/\wp$ admits a reduced representative with all components in $x^{-1}k[x^{-1}]$, produced by induction on r through the truncation $W_r \twoheadrightarrow W_{r-1}$, using that adding $V^{r-1}(h)$ is carry-free. (The “principal part” step of the earlier note is *not* used: componentwise truncation of a Witt vector is not a legitimate operation.) Gap 2 is closed by the Witt-coordinate analogue of the thesis’s invariant-ring computation: the Witt sum $\vec{Y} = \bigoplus_i \vec{X}_i$ is invariant under the twisted G^{d-1} -action and under S_d , satisfies $\wp \vec{Y} = \bigoplus_i \vec{f}(x_i)$, and the resulting map of G -torsors is automatically an isomorphism. Together with the (verified) degree count of the earlier note, reproduced here for completeness, this yields an unconditional proof of the theorem, directly for all r at once. The single external input remains the Artin–Schreier–Witt break formula, stated here in the precise form and direction used.

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1 Setting, conventions, and what is being proved

Throughout, k is an algebraically closed field of characteristic $p > 0$ and $K = k((x))$. We prove:

Theorem 1 (main theorem; theorem:ramification_after_blowup_is_tame_for_p-power). *Let C be a projective smooth geometrically connected curve over k , let $G = \mathbb{Z}/p^r\mathbb{Z}$, and let $\mathcal{P} \rightarrow U = C \setminus \mathfrak{m}$ be a G -torsor with ramification bounded by $\mathfrak{m}$. Let $\mathfrak{n} = nP \subset \mathfrak{m}$ and $d := \deg \mathfrak{n} = n$. Then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*

Here, as in the thesis, “ramification bounded by d at P ” means that the d -th *upper-numbering* ramification group of the local extension at P is trivial (**definition:local_ramification_boundness**); equivalently, the largest upper-numbering break u of that extension satisfies $u < d$ *strictly*.

Standing reduction. As in the current (partial) proof in the thesis, we may assume that k is algebraically closed and that $\mathcal{P} \rightarrow U$ is *totally ramified* at P with inertia equal to all of G : if the image H of inertia at P is a proper subgroup, decompose $\mathcal{P} \rightarrow \mathcal{P}_{G/H} \rightarrow U$; then $\mathcal{P}_{G/H}$ is unramified at P , extends over P , and is locally trivial at any point over P (as $k = \bar{k}$), which replaces $(G, \mathcal{P})$ by $(H, \mathcal{P} \rightarrow \mathcal{P}_{G/H})$ with $H \cong \mathbb{Z}/p^s\mathbb{Z}$. This reduction is valid for all r and is kept unchanged; we do not repeat it below. After it, the restriction $\mathcal{Q} := \mathcal{P}_x$ of $\mathcal{P}$ to the punctured formal disc $\text{Spec } k((x))$ at P (with x a uniformizer at P) is a *connected* $\mathbb{Z}/p^r\mathbb{Z}$ -torsor, i.e. $\mathcal{Q} = \text{Spec } L$ for a cyclic, totally ramified extension L/K of degree p^r , with ramification bounded by d .

Witt vector conventions. W_r denotes p -typical Witt vectors of length r , with components indexed $0, \dots, r-1$; $\oplus, \ominus$ denote Witt addition and subtraction, given by the universal polynomials S_n : the n -th component of $\vec{a} \oplus \vec{b}$ is

$$(\vec{a} \oplus \vec{b})_n = a_n + b_n + C_n(a_0, \dots, a_{n-1}; b_0, \dots, b_{n-1}) \quad (1)$$

for a universal polynomial C_n over $\mathbb{Z}$ in the *lower* components only. On $\mathbb{F}_p$ -algebras the Frobenius F is componentwise: $F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p)$; we set $\wp := F - \text{id}$ (computed with $\ominus$), so for $r = 1$, $\wp(u) = u^p - u$. The Verschiebung is $V(a_0, \dots, a_{r-2}) = (0, a_0, \dots, a_{r-2})$, and $t: W_r \rightarrow W_{r-1}$, $t(a_0, \dots, a_{r-1}) = (a_0, \dots, a_{r-2})$, is the truncation. W_r is a functor: a ring homomorphism $\sigma: A \rightarrow B$ induces $W_r(\sigma): W_r(A) \rightarrow W_r(B)$ acting *componentwise*, $W_r(\sigma)(a_0, \dots, a_{r-1}) = (\sigma a_0, \dots, \sigma a_{r-1})$, and this is a ring homomorphism commuting with F , V , t , hence with $\wp$. In particular a ring automorphism σ of A commutes with $\oplus$ and $\ominus$, and a Witt vector $\vec{a} \in W_r(A)$ satisfies $W_r(\sigma)(\vec{a}) = \vec{a}$ if and only if each component a_n is σ -invariant. By Artin–Schreier–Witt (ASW) theory, over a field $M \supseteq \mathbb{F}_p$ one has $H^1(M, \mathbb{Z}/p^r\mathbb{Z}) \cong W_r(M)/\wp W_r(M)$, the class of $\vec{g}$ corresponding to the torsor $\text{Spec } M[\vec{X}]/(\wp \vec{X} \ominus \vec{g})$ (notation of Lemma 3 below); connected torsors correspond to classes of order p^r , and the extension is then cyclic of degree p^r , totally ramified when $M = k((x))$ with k algebraically closed.

We record the elementary Witt bookkeeping used repeatedly.

Lemma 2. *Let A be an $\mathbb{F}_p$ -algebra.*

(i) $t: W_r(A) \rightarrow W_{r-1}(A)$ is a ring homomorphism commuting with F , hence with $\wp$; its kernel is $V^{r-1}A = \{(0, \dots, 0, c) : c \in A\}$.

(ii) $V^{r-1}: A \rightarrow W_r(A)$ is additive, and $\wp \circ V^{r-1} = V^{r-1} \circ \wp$.

(iii) (*carry-free top-level addition*) For all $\vec{a} \in W_r(A)$ and $c \in A$,

$$\vec{a} \oplus V^{r-1}(c) = (a_0, \dots, a_{r-2}, a_{r-1} + c).$$

Proof. (i) is standard (t is a natural transformation of ring functors; F is componentwise in characteristic p), and the kernel is computed componentwise. (ii) Additivity of V is standard; since F is componentwise it commutes with V on $\mathbb{F}_p$ -algebras, hence so does $\wp = F - \text{id}$. (iii) This is a universal identity, so it suffices to prove it over $\mathbb{Z}[a_0, \dots, a_{r-1}, c]$, where the ghost components $w_n(\vec{a}) = \sum_{j \leq n} p^j a_j^{p^{n-j}}$ are injective jointly with torsion-freeness. Put $\vec{b} = (0, \dots, 0, c)$ and

$\vec{s} = \vec{a} \oplus \vec{b}$. For $n < r - 1$ we have $w_n(\vec{b}) = 0$, so $w_n(\vec{s}) = w_n(\vec{a})$, and by induction on n (solving for s_n using torsion-freeness) $s_n = a_n$ for $n \leq r - 2$. For $n = r - 1$, $w_{r-1}(\vec{b}) = p^{r-1}c$ gives $p^{r-1}s_{r-1} = p^{r-1}a_{r-1} + p^{r-1}c$, i.e. $s_{r-1} = a_{r-1} + c$. $\square$

Lemma 3 (ASW covers are étale torsors). *Let A be an $\mathbb{F}_p$ -algebra and $\vec{g} \in W_r(A)$. Let*

$$B := A[X_0, \dots, X_{r-1}] / (\text{the } r \text{ components of } \wp(\vec{X}) \ominus \vec{g}), \quad \vec{X} = (X_0, \dots, X_{r-1}).$$

Then B is finite free over A with basis $\{\prod_n X_n^{a_n} : 0 \leq a_n \leq p-1\}$, and étale over A . The translation action of $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$, $\vec{X} \mapsto \vec{X} \oplus \underline{m}$ (where $\underline{m}$ is the image of m under $\mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p)$), makes $\text{Spec } B$ a $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\text{Spec } A$.

Proof. By (1) and componentwise F , the n -th defining equation has the triangular form

$$X_n^p - X_n - g_n + C'_n(X_0, \dots, X_{n-1}; g_0, \dots, g_{n-1}) = 0$$

for universal polynomials C'_n . Triangularity gives the free basis; the Jacobian matrix of the system is lower triangular with diagonal entries $pX_n^{p-1} - 1 = -1$, a unit, so B is étale over A . Finally, $\text{Spec } B$ is the pullback along $\vec{g}: \text{Spec } A \rightarrow \mathbb{W}_r$ of the Lang covering $\wp: \mathbb{W}_r \rightarrow \mathbb{W}_r$ of the Witt group scheme over $\mathbb{F}_p$, which is a surjective étale group homomorphism with kernel $W_r(\mathbb{F}_p)$; hence $\text{Spec } B$ is a $W_r(\mathbb{F}_p)$ -torsor. $\square$

Lemma 4. *Let G be a finite group scheme over a base T and let $\phi: \mathcal{P}_1 \rightarrow \mathcal{P}_2$ be a G -equivariant T -morphism of G -torsors (for the fppf or étale topology). Then ϕ is an isomorphism.*

Proof. Locally on a cover trivializing both torsors, ϕ is a G -equivariant endomorphism of the trivial torsor G_T , i.e. right translation by a section of G , hence an isomorphism; being an isomorphism descends. $\square$

2 Gap 1: reduced polynomial representatives and realization over

$\mathbb{G}_m$

The earlier note took $\vec{f}$ to be “the principal part” of the Witt datum. As the review observes, this is not a legitimate operation (Witt addition is not componentwise, so componentwise truncation to polar parts changes the ASW class), and in any case no W_r -realization statement was available in the thesis, whose realization lemma (`lemma:realization_over_Gm`) covers only the tame case and $G = \mathbb{Z}/p\mathbb{Z}$. This section proves the correct statement.

Definition 5. For $g \in k((x))$ put $m(g) := \max(0, -v_x(g))$ (the pole order, or 0 if g is regular). A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(k((x)))$ is *reduced* if for every j , either $m(g_j) = 0$ or $p \nmid m(g_j)$. It is a *reduced polynomial vector* if moreover every $g_j \in x^{-1}k[x^{-1}]$ (so that $m(g_j) = \deg_{x^{-1}} g_j$, and $m(g_j) = 0$ if and only if $g_j = 0$).

Lemma 6 (surjectivity of $\wp$ on integral Witt vectors). *$\wp: W_r(k[[x]]) \rightarrow W_r(k[[x]])$ is surjective.*

Proof. Induction on r . For $r = 1$ this is proved in the thesis (`lemma:realization_over_Gm`, first bullet): given $h = \sum_{i \geq 0} b_i x^i$, solve $u^p - u = h$ coefficientwise, using that k is algebraically closed for the constant term. For $r > 1$, let $\vec{h} \in W_r(k[[x]])$. By induction choose $\vec{u}' \in W_{r-1}(k[[x]])$ with $\wp \vec{u}' = t(\vec{h})$, and lift it to $\tilde{u} := (\vec{u}', 0) \in W_r(k[[x]])$. By Lemma 2(i), $t(\wp \tilde{u} \ominus \vec{h}) = \wp \vec{u}' \ominus t(\vec{h}) = 0$, so $\wp \tilde{u} \ominus \vec{h} = V^{r-1}(g)$ for some $g \in k[[x]]$. Choose $w \in k[[x]]$ with $\wp w = -g$ (case $r = 1$ again). Then by Lemma 2(ii) and additivity of V^{r-1} ,

$$\wp(\tilde{u} \oplus V^{r-1}(w)) = \wp \tilde{u} \oplus V^{r-1}(\wp w) = (\vec{h} \oplus V^{r-1}(g)) \oplus V^{r-1}(-g) = \vec{h}. \quad \square$$

Proposition 7 (reduced polynomial representatives). *Every class in $W_r(k((x)))/\wp W_r(k((x)))$ has a reduced polynomial representative: there exists $\vec{u} \in W_r(k((x)))$ such that $\vec{g} := \vec{f} \ominus \wp \vec{u}$ has all components in $x^{-1}k[x^{-1}]$ with pole orders either 0 or prime to p .*

Proof. Induction on r .

Base case $r = 1$. This is the argument of `lemma:realization_over_Gm(2)` in the thesis: first, writing $f = f^- + f^+$ with $f^- \in x^{-1}k[x^{-1}]$ and $f^+ \in k[[x]]$, surjectivity of $\wp$ on $k[[x]]$ removes f^+ ; then, as long as the leading term is cx^{-m} with $p \mid m$, $m > 0$, subtracting $\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}$ strictly decreases the pole order while staying inside $x^{-1}k[x^{-1}]$; this terminates at 0 or at pole order prime to p .

Inductive step. Let $\vec{f} \in W_r(k((x)))$. By induction there is $\vec{u}' \in W_{r-1}(k((x)))$ such that $\vec{g}' := t(\vec{f}) \ominus \wp \vec{u}'$ is a reduced polynomial vector in W_{r-1} . Lift $\vec{u}'$ to $\tilde{u} := (\vec{u}', 0) \in W_r(k((x)))$ and set $\vec{h} := \vec{f} \ominus \wp \tilde{u}$. By Lemma 2(i), $t(\vec{h}) = \vec{g}'$, i.e.

$$\vec{h} = (g'_0, \dots, g'_{r-2}, h_{r-1}), \quad h_{r-1} \in k((x)),$$

with the first $r-1$ components already reduced polynomials. It remains to repair the last component *without touching the others*; by Lemma 2(iii), adjustments of the form $\ominus \wp(V^{r-1}(z)) = \ominus V^{r-1}(\wp z)$ change only the last component, by $-\wp(z)$. So:

- Write $h_{r-1} = h^- + h^+$ with $h^- \in x^{-1}k[x^{-1}]$, $h^+ \in k[[x]]$, choose $w \in k[[x]]$ with $\wp w = h^+$ (Lemma 6 with $r = 1$), and replace $\vec{h}$ by $\vec{h} \ominus \wp(V^{r-1}w)$: the last component becomes h^- .
- While the last component has leading term cx^{-m} with $p \mid m$, $m > 0$, replace $\vec{h}$ by $\vec{h} \ominus \wp(V^{r-1}(c^{1/p}x^{-m/p}))$: the last component changes by $-\wp(c^{1/p}x^{-m/p}) = -cx^{-m} + c^{1/p}x^{-m/p}$, so its pole order strictly decreases and it stays in $x^{-1}k[x^{-1}]$. This terminates at 0 or at pole order prime to p .

The result is a reduced polynomial vector $\wp$ -equivalent to $\vec{f}$. □

The break formula. The following is the single external input. It is due, over finite residue fields, to Kanesaka–Sekiguchi [2] and Thomas [3] (building on Schmid [1] for $r = 1$), and over arbitrary perfect residue fields to Elder–Keating [4].

Proposition 8 (Schmid–Witt break formula, direction used). *Let $K = k((x))$ with k perfect of characteristic p , and let L/K be a cyclic totally ramified extension of degree p^r whose ASW class is represented by a reduced vector $\vec{g} \in W_r(K)$ (Definition 5), with $m_j := m(g_j)$. Then the largest upper-numbering ramification break of L/K equals*

$$u = \max_{0 \leq j \leq r-1} p^{r-1-j} m_j.$$

For $r = 1$ this is the classical statement (break = m_0) already used in the thesis.

Note that reducedness is exactly what makes the formula exact: two nonzero terms $p^{r-1-j} m_j = p^{r-1-j'} m_{j'}$ with $j < j'$ would force $p \mid m_{j'}$, so the maximum is achieved at a unique level and no cancellation of breaks can occur. We use only the stated direction: *evaluating the formula on one reduced representative computes u* ; no minimization over representatives is needed.

Lemma 9 (realization over $\mathbb{G}_m$ for W_r). *Let k be algebraically closed of characteristic $p > 0$ and let $\mathcal{Q}$ be a connected $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\text{Spec}(k((x)))$ with ramification bounded by d (`definition:local_ramification`). Then there is a reduced polynomial vector $\vec{f} = (f_0, \dots, f_{r-1})$, $f_j \in x^{-1}k[x^{-1}]$, such that, with $m_j := \deg_{x^{-1}} f_j$:*

1. $p^{r-1-j} m_j < d$ for all $0 \leq j \leq r-1$; equivalently $m_j < d p^{j-(r-1)}$;
2. $\mathcal{P}' := \text{Spec}\left(k[x, x^{-1}][X_0, \dots, X_{r-1}]/(\wp \vec{X} \ominus \vec{f})\right)$ is a $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$;
3. $\mathcal{P}'$ extends to a torsor over $\mathbb{P}_k^1 \setminus \{0\}$ — in particular it is unramified at ∞ — and has ramification bounded by d at 0.

Proof. $\mathcal{Q} = \text{Spec } L$ with L/K cyclic totally ramified of degree p^r and largest upper break $u < d$. Let $[\vec{f}_0] \in W_r(K)/\wp$ be its ASW class and let $\vec{f}$ be a reduced polynomial representative (Proposition 7).

First, $f_0 \neq 0$: if $f_0 = 0$ then $\vec{f} = V(\vec{b})$ for some $\vec{b} \in W_{r-1}(K)$, and $p^{r-1}\vec{f} = (VF)^{r-1}V\vec{b} = V^r F^{r-1}\vec{b} = 0$ in $W_r(K)$, so the class has order $\leq p^{r-1}$, contradicting $[L : K] = p^r$. Hence $m_0 \geq 1$ and $p \nmid m_0$, and Proposition 8 applies: $\max_j p^{r-1-j} m_j = u < d$, which is 1.

For 2: by Lemma 3 (with $A = k[x, x^{-1}] \ni f_j$), $\mathcal{P}'$ is a $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\mathbb{G}_m$; its restriction to $\text{Spec } k((x))$ is the ASW torsor of the class of $\vec{f}$, i.e. $\mathcal{Q}$.

For 3: in the coordinate $y = x^{-1}$ at ∞ we have $f_j \in y k[y]$, so $\vec{f} \in W_r(k[y])$ and Lemma 3 (with $A = k[y]$) extends $\mathcal{P}'$ to a torsor over $\mathbb{A}_k^1 = \text{Spec } k[y] = \mathbb{P}_k^1 \setminus \{0\}$; a torsor over the full local ring at ∞ is unramified there. The ramification bound at 0 holds because $\mathcal{P}'_x \cong \mathcal{Q}$. $\square$

Remark 10. Connectedness of $\mathcal{Q}$ is guaranteed in the application by the standing reduction of §1 (inertia equal to all of G). This replaces — and corrects — the “take the principal part” step: the components f_j are honest polynomials in x^{-1} , so substituting $x \mapsto x_i$ and forming the Witt sum below are purely algebraic operations, and all subsequent $\wp$ -adjustments were already performed *before* symmetrization, inside $W_r(k((x)))$.

Remark 11 (sanity check). For every j with $p^{r-1-j} \geq d$, the bound 1 forces $m_j = 0$, i.e. $f_j = 0$: all sufficiently low Witt components of a reduced representative vanish outright. This makes visible how strong the hypothesis $u < d$ is at the lower levels.

3 Gap 2: the symmetric-power cover at η is the ASW cover of the Witt sum

We now generalize the invariant-ring computation of the thesis’s $r = 1$ proof (theorem:ramification_after_blowup) Artin–Schreier case) to Witt coordinates. Fix $\vec{f} \in W_r(R)$, $R = k[x, x^{-1}]$, and let

$$S = R[\vec{X}]/(\wp \vec{X} \ominus \vec{f})$$

be the coordinate ring of the torsor $\mathcal{P}' \rightarrow \mathbb{G}_m$ of Lemma 9. Write $R^{\otimes_k d} = k[x_1^{\pm 1}, \dots, x_d^{\pm 1}]$ and

$$S^{\otimes_k d} = R^{\otimes_k d}[\vec{X}_1, \dots, \vec{X}_d]/(\wp \vec{X}_1 \ominus \vec{f}(x_1), \dots, \wp \vec{X}_d \ominus \vec{f}(x_d)),$$

where $\vec{X}_i = (X_{i,0}, \dots, X_{i,r-1})$ and $\vec{f}(x_i)$ is the image of $\vec{f}$ under $x \mapsto x_i$. Exactly as in the thesis, the d -fold product torsor $p_1^{-1}\mathcal{P}' \otimes \dots \otimes p_d^{-1}\mathcal{P}'$ on U^d is by definition the quotient of $\text{Spec } S^{\otimes_k d}$ by G^{d-1} , embedded in G^d as the kernel of the sum map, and acting (in Witt coordinates, writing $\vec{g}$ for the image of $g \in G$ under $\mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p)$) by

$$(g_1, \dots, g_{d-1}): \quad \vec{X}_1 \mapsto \vec{X}_1 \oplus \vec{g}_1, \quad \vec{X}_i \mapsto \vec{X}_i \oplus \vec{g}_i \ominus \vec{g}_{i-1} \ (1 < i < d), \quad \vec{X}_d \mapsto \vec{X}_d \ominus \vec{g}_{d-1}.$$

Proposition 12 (identification of the product torsor). *Set*

$$\vec{Y} := \bigoplus_{i=1}^d \vec{X}_i \in W_r(S^{\otimes_k d}), \quad \vec{\alpha} := \bigoplus_{i=1}^d \vec{f}(x_i) \in W_r(R^{\otimes_k d}),$$

where $\bigoplus$ denotes the d -fold Witt vector sum $\oplus$ of (1) (not a direct sum); in particular $\vec{Y}$ is a single Witt vector of length r . Then:

1. Every component Y_n of $\vec{Y}$ is G^{d-1} -invariant, and

$$(S^{\otimes_k d})^{G^{d-1}} \cong R^{\otimes_k d}[\vec{Y}] / (\wp \vec{Y} \ominus \vec{\alpha})$$

as G -torsors over $R^{\otimes_k d}$ (the residual $G = G^d / G^{d-1}$ acting by $\vec{Y} \mapsto \vec{Y} \oplus \vec{g}$).

2. Every component α_n of $\vec{\alpha}$ is a symmetric Laurent polynomial, i.e. lies in $(R^{\otimes_k d})^{S_d} = k[e_1, \dots, e_d, e_d^{-1}]$, and

$$\left((S^{\otimes_k d})^{G^{d-1}} \right)^{S_d} \cong k[e_1, \dots, e_d, e_d^{-1}][\vec{Y}] / (\wp \vec{Y} \ominus \vec{\alpha}).$$

Consequently, restricting to the complete local field $\widehat{K}_\eta = \kappa(\eta)((e_d))$, $\kappa(\eta) = k(e_1/e_d, \dots, e_{d-1}/e_d)$, of `eq:complete_field_at_generic_point`,

$$\mathcal{P}'^{[d]}|_{\text{Spec } \widehat{K}_\eta} \cong \text{Spec } \widehat{K}_\eta[\vec{Z}] / (\wp \vec{Z} \ominus \vec{\alpha}).$$

Proof. 1 An element $\sigma = (g_1, \dots, g_{d-1}) \in G^{d-1}$ acts as a ring automorphism of $S^{\otimes_k d}$: this is its *original* action, defined on the generators $X_{i,n}$ by the twisted formulas above (which mix components through the carry polynomials of (1)). By functoriality of W_r (see the conventions), the induced map $W_r(\sigma)$ on $W_r(S^{\otimes_k d})$ is componentwise and commutes with $\oplus, \ominus$; on the generators it reproduces the twisted action, $W_r(\sigma)(\vec{X}_i) = \vec{X}_i \oplus \vec{g}_i \ominus \vec{g}_{i-1}$ (with $\vec{g}_0 = \vec{g}_d = \vec{0}$). Hence, using commutativity and associativity of $\oplus$, the Witt sum transforms by the telescoping constant

$$W_r(\sigma)(\vec{Y}) = \bigoplus_{i=1}^d W_r(\sigma)(\vec{X}_i) = \vec{Y} \oplus \vec{g}_1 \oplus (\vec{g}_2 \ominus \vec{g}_1) \oplus \dots \oplus (\ominus \vec{g}_{d-1}) = \vec{Y}.$$

Since $W_r(\sigma)$ is componentwise, this equality of r -tuples says exactly $\sigma(Y_n) = Y_n$ for every n , i.e. each $Y_n \in S^{\otimes_k d}$ is invariant. Since $\wp = F - \text{id}$ is additive (F is a ring endomorphism),

$$\wp \vec{Y} = \bigoplus_{i=1}^d \wp \vec{X}_i = \bigoplus_{i=1}^d \vec{f}(x_i) = \vec{\alpha}.$$

Hence there is a well-defined $R^{\otimes_k d}$ -algebra homomorphism

$$\phi: T := R^{\otimes_k d}[\vec{Z}] / (\wp \vec{Z} \ominus \vec{\alpha}) \longrightarrow (S^{\otimes_k d})^{G^{d-1}}, \quad \vec{Z} \mapsto \vec{Y}.$$

By Lemma 3, $\text{Spec } T$ is a G -torsor over U^d . The target is the product torsor (the contracted product of the $p_i^{-1}\mathcal{P}'$ along the sum map $G^d \rightarrow G$), a G -torsor over U^d by the thesis's construction of torsor products; its residual G -action is induced by $(g, 0, \dots, 0) \in G^d$, i.e. $\vec{X}_1 \mapsto \vec{X}_1 \oplus \vec{g}$, under which $\vec{Y} \mapsto \vec{Y} \oplus \vec{g}$. So $\text{Spec } \phi$ is a G -equivariant morphism of G -torsors over U^d , hence an isomorphism by Lemma 4.

2 S_d acts on $S^{\otimes_k d}$ by permuting the tensor factors, i.e. $x_i \mapsto x_{\sigma(i)}$, $\vec{X}_i \mapsto \vec{X}_{\sigma(i)}$. As in 1, each $\sigma \in S_d$ is a ring automorphism, so $W_r(\sigma)$ is componentwise and commutes with $\oplus$. Since $\oplus$ is commutative and associative, permuting the summands of a Witt sum leaves it unchanged; thus $\vec{Y}$ is S_d -invariant (componentwise, as before) and $W_r(\sigma)(\vec{\alpha}) = \bigoplus_i \vec{f}(x_{\sigma(i)}) = \vec{\alpha}$, so each $\alpha_n \in R^{\otimes_k d}$ is a symmetric Laurent polynomial, hence lies in $k[e_1, \dots, e_d, e_d^{-1}]$ (the S_d -invariants of $R^{\otimes_k d}$, as in the thesis). By Lemma 3, T is free over $R^{\otimes_k d}$ with basis the monomials $\prod_n Y_n^{a_n}$, $0 \leq a_n < p$, each of which is S_d -invariant; writing an element in this basis, it is S_d -invariant if and only if all its coefficients are. Therefore

$$T^{S_d} = (R^{\otimes_k d})^{S_d}[\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha}) = k[e_1, \dots, e_d, e_d^{-1}][\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha}).$$

The final statement follows exactly as in the $r = 1$ case: by definition $\mathcal{P}'^{[d]}$ is the $\Xi = G^{d-1} \rtimes S_d$ -quotient above, and its restriction to $\text{Spec } \widehat{K}_\eta$ along $k[e_1, \dots, e_d, e_d^{-1}] \rightarrow \widehat{K}_\eta$ is the ASW cover of the image of $\vec{\alpha}$ in $W_r(\widehat{K}_\eta)$. $\square$

Remark 13. The proof via Lemma 4 replaces the rank count of the $r = 1$ argument ($p^{rd}/p^{r(d-1)} = p^r$ together with irreducibility of the invariant's minimal equation); the rank count alone would still require showing that $\vec{Y}$ generates the invariant ring, which is exactly what equivariance gives for free. Note also that a pure restriction argument on H^1 's ("the class of the product torsor is the sum of the pulled-back characters") identifies the cover over $k(x_1, \dots, x_d)$ but does *not* by itself descend it to the symmetric base $k(e_1, \dots, e_d)$; the invariant-theoretic computation above is what effects the descent.

4 The degree count

This section reproduces, for completeness, the two elementary lemmas and the estimate of the earlier note (verified line by line in the review), now fed with the legitimate inputs provided by Lemma 9 and Proposition 12. Throughout, $\vec{f}$ is the reduced polynomial vector of Lemma 9, $m_j = \deg_{x^{-1}} f_j$ satisfies $m_j < dp^{j-(r-1)}$, and $y_i := 1/x_i$, so that $f_j(x_i) \in y_i k[y_i]$ is a polynomial in y_i of degree m_j .

Lemma 14 (symmetric regularity). *Let $\Phi \in k[y_1, \dots, y_d]^{S_d}$ be symmetric with every monomial of total degree $< d$. Then $\Phi \in \kappa(\eta) = k(e_1/e_d, \dots, e_{d-1}/e_d)$; in particular Φ is regular at η .*

Proof. Φ is a k -combination of monomial symmetric functions $m_\lambda(y)$ with $|\lambda| < d$. Each $m_\lambda(y)$ is an isobaric polynomial in the elementary symmetric functions $e_\mu(y) = \prod_i e_{\mu_i}(y)$ with $\mu \vdash |\lambda|$; every part $\mu_i \leq |\lambda| < d$, so only $e_1(y), \dots, e_{d-1}(y)$ occur — never $e_d(y)$. Now $e_k(y) = e_{d-k}(x)/e_d(x)$, which for $1 \leq k \leq d-1$ lies in $\kappa(\eta)$. Hence $\Phi \in \kappa(\eta) \subset \widehat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$. $\square$

Lemma 15 (isobaricity of Witt addition). *Give the variable $f_j(x_i)$ the weight p^j . Then each component $\alpha_n = (\bigoplus_{i=1}^d \vec{f}(x_i))_n$ is isobaric of weight p^n : every monomial $\prod_{i,j} (f_j(x_i))^{e_{ij}}$ of α_n satisfies $\sum_{i,j} e_{ij} p^j = p^n$.*

Proof. The n -th ghost component $w_n(\vec{a}) = \sum_{j \leq n} p^j a_j^{p^{n-j}}$ is isobaric of weight p^n when $\deg a_j = p^j$. The universal addition polynomials S_n are determined over $\mathbb{Z}$ by additivity of the ghost components and the recursion $p^n S_n = \sum_j p^j (a_j^{p^{n-j}} + b_j^{p^{n-j}}) - \sum_{j < n} p^j S_j^{p^{n-j}}$, which preserves isobaricity of weight p^n by induction; the same holds after reduction mod p and under d -fold iteration of $\oplus$. Hence α_n , a universal isobaric polynomial evaluated at the $f_j(x_i)$, is isobaric of weight p^n . $\square$

Theorem 16 (regularity of all Witt components). *With $\vec{f}$ as in Lemma 9, every component α_n ($0 \leq n \leq r-1$) of $\vec{\alpha} = \bigoplus_{i=1}^d \vec{f}(x_i)$ lies in $\kappa(\eta)$; in particular $\vec{\alpha} \in W_r(\hat{\mathcal{O}}_\eta)$.*

Proof. Fix $n \leq r-1$ and a monomial $\prod_{i,j} (f_j(x_i))^{e_{ij}}$ of α_n ; set $E_j := \sum_i e_{ij}$, so $\sum_j E_j p^j = p^n$ by Lemma 15. Expanding each $f_j(x_i)$ as a polynomial in y_i of degree m_j , every resulting monomial in the y_i has total degree

$$\leq \sum_{i,j} e_{ij} m_j = \sum_j E_j m_j < \sum_j E_j d p^{j-(r-1)} = d p^{-(r-1)} \sum_j E_j p^j = d p^{n-(r-1)} \leq d,$$

strictly: some $E_j > 0$ (as $\sum_j E_j p^j = p^n > 0$), and the pole bound $m_j < d p^{j-(r-1)}$ of Lemma 91 is strict even when $m_j = 0$. Thus every monomial of α_n — a symmetric polynomial in $y_1, \dots, y_d$ by Proposition 122 — has total y -degree $< d$, and Lemma 14 gives $\alpha_n \in \kappa(\eta)$. $\square$

5 Proof of the theorem

Proof of Theorem 1. By the standing reduction of §1 we may assume k algebraically closed and $\mathcal{P} \rightarrow U$ totally ramified at P with inertia all of G , so that $\mathcal{Q} := \mathcal{P}_x$ is a connected $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\text{Spec } k((x))$ with ramification bounded by $d = \deg \mathbf{n}$.

Lemma 9 produces a torsor $\mathcal{P}' \rightarrow \mathbb{G}_m$, defined by a reduced polynomial vector $\vec{f}$ with $p^{r-1-j} m_j < d$, with $\mathcal{P}'_x \cong \mathcal{Q}$. By the reduction to $\mathbb{P}^1$ (`cor:reduction_to_P1`, valid for arbitrary finite abelian G), the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathbf{n}}$ agrees with that of $\mathcal{P}'^{[d]}$ at $\eta_{d \cdot 0}$; so we may assume $C = \mathbb{P}_k^1$, $\mathbf{m} = d \cdot 0 + \infty$, $\mathbf{n} = d \cdot 0$, $\mathcal{P} = \mathcal{P}'$.

By Proposition 12,

$$\mathcal{P}'^{[d]}|_{\text{Spec } \hat{K}_\eta} \cong \text{Spec } \hat{K}_\eta[\vec{Z}] / (\wp \vec{Z} \ominus \vec{\alpha}), \quad \vec{\alpha} = \bigoplus_{i=1}^d \vec{f}(x_i),$$

with $\hat{K}_\eta = \kappa(\eta)((e_d))$ the complete local field at $\eta_{\mathbf{n}}$ and $\hat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$ its valuation ring.

By Theorem 16, $\vec{\alpha} \in W_r(\kappa(\eta)) \subset W_r(\hat{\mathcal{O}}_\eta)$. Hence, by Lemma 3 applied over $A = \hat{\mathcal{O}}_\eta$, the ASW cover $\wp \vec{Z} = \vec{\alpha}$ is finite étale over $\hat{\mathcal{O}}_\eta$, and $\mathcal{P}'^{[d]}|_{\text{Spec } \hat{K}_\eta}$ is its generic fibre: every field factor of $\hat{K}_\eta[\vec{Z}] / (\wp \vec{Z} \ominus \vec{\alpha})$ is the fraction field of a finite étale $\hat{\mathcal{O}}_\eta$ -algebra, i.e. an unramified extension of $\hat{K}_\eta$ with respect to the discrete valuation ring $\hat{\mathcal{O}}_\eta$. By `definition:tamely_ramified_in_codim_1`, $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathbf{n}}$. $\square$

Remark 17 (this is a direct proof, not an induction). The argument proves all r simultaneously; it reproves the case $r = 1$ (`cor:ramification_after_blowup_is_tame(1)`) as the special case of a vector of length one and never invokes it. In particular, when merging into the thesis, the current proof body of `theorem:ramification_after_blowup_is_tame_for_p_power` from the sentence “We proceed by induction on r ” onward should be *replaced* by §§2–5 above — including the step which applies the $r = 1$ theorem to the top layer $\mathcal{P} \rightarrow \mathcal{P}'$: that step is unjustified, since the hypothesis bounds the ramification of $\mathcal{P}$ in *upper* numbering, while the break of the top degree- p layer is the largest *lower*-numbering break, which can be of size about $p^{r-1}d$. The reduction paragraph preceding it (algebraically closed base, totally ramified, inertia all of G) is correct for all r and is exactly the standing reduction of §1.

Remark 18 (bookkeeping of the estimate). As in the earlier note, the bound is exact: the weight $p^{j-(r-1)}$ is paired against the isobaricity constraint $\sum_j E_j p^j = p^n$, so the estimate $d p^{n-(r-1)} \leq d$

is independent of how the Witt carries distribute the exponents E_j ; the hypothesis “break $< d$ at p_∞ ” is used at *every* Witt level at once. Remark 11 records the extreme case: components with $p^{r-1-j} \geq d$ vanish outright in the reduced representative.

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